Levitaz vs Chubanga

I. Introduction

A. Context

This document presents an analysis of straight-line sailing exercises performed simultaneously by Gian Stragiotti and Max Maeder on November 25, 26, and 27, 2025. During these sessions, the riders alternated foil masts between Levitaz and Chubanga, allowing for direct performance comparisons under closely matched conditions.

The objective of this study is to assess the performance impact of the foil mast while controlling as much as possible for environmental and configuration-related factors.

B. Methodology

The python analysis is available on *LeviVSChub.html*. It relies on pre-extracted sailing intervals, ensuring that both riders were exposed to the same wind and sea-state conditions during each comparison. Full details on the interval extraction process and the content of the analyzed runs are provided in the files *interval_selection.html* and *report.html.*

Two complementary analytical approaches are explored:

- **Paired-difference analysis.**
  The first approach focuses on the performance difference between the two foil masts by directly analyzing, for each interval, the difference in speed over ground (SOG) between Levitaz and Chubanga. This paired structure allows for a clean comparison under matched conditions.
- **Rider-specific analysis.**
  The second approach separates the observations by rider and examines performance when sailing with the Levitaz mast versus the Chubanga mast. This perspective allows rider-specific responses to the foil masts to be examined explicitly.

For each approach, the analysis proceeds in two steps. First, an exploratory analysis based on mean comparisons and t-tests is conducted to provide an initial, descriptive view of the differences. Second, ordinary least squares (OLS) regressions are applied to quantify the effect of the foil mast while controlling for relevant sailing conditions and configurations.

II. Results

A. Approach I: Paired-Difference Analysis

This first approach focuses on paired comparisons between the Levitaz and Chubanga foil masts. For each extracted interval, the difference in speed over ground is computed as SOG_Levi − SOG_Chub, ensuring that both riders are compared under identical wind and sea-state conditions.

1. Descriptive statistics and exploratory analysis

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Leg-type exposure by mast:
- Chub: 27 intervals (48.214%) were downwind.
- Chub: 29 intervals (51.786%) were upwind.
- Levi: 27 intervals (48.214%) were downwind.
- Levi: 29 intervals (51.786%) were upwind.

Rider-mast exposure:
- Gian Stragiotti sailed 31 intervals on Chub (55.4%) and 25 on Levi (44.643%).
- Max Maeder sailed 25 intervals on Chub (44.6%) and 31 on Levi (55.357%).

======================================================================
All legs
Average conditions – TWS: 16.576/16.577 kn, TWA: -60.538/-60.911°, Weight: 111.812/111.788 kg (Levi / Chub).
Mean SOG difference (Levi - Chub): -0.142 kn. Chub is faster on average in the raw data.
According to the Welch t-test on avg_SOG: t = -0.15, p = 0.885
The difference in mean SOG is not statistically significant (p = 0.885 ≥ 0.05).
======================================================================
Upwind only
Average conditions – TWS: 16.874/16.874 kn, TWA: 30.229/29.318°, Weight: 111.814/111.786 kg (Levi / Chub).
Mean SOG difference (Levi - Chub): 0.107 kn. Levi is faster on average in the raw data.
According to the Welch t-test on avg_SOG: t = 0.21, p = 0.831
The difference in mean SOG is not statistically significant (p = 0.831 ≥ 0.05).
======================================================================
Downwind only
Average conditions – TWS: 16.257/16.258 kn, TWA: -158.028/-157.824°, Weight: 111.811/111.789 kg (Levi / Chub).
Mean SOG difference (Levi - Chub): -0.409 kn. Chub is faster on average in the raw data.
According to the Welch t-test on avg_SOG: t = -1.25, p = 0.217
The difference in mean SOG is not statistically significant (p = 0.217 ≥ 0.05).
```

Exposure is well balanced across sailing regimes and riders. For both masts, approximately half of the intervals are upwind and half are downwind, and both riders sail a comparable number of intervals on each mast.

Across all legs, sailing conditions are identical between Levitaz and Chubanga: this is the interest of working on paired difference, as both masts are compared exactly at the same time, with the same conditions. Gian's weight variations (112.1 / 112.2) and Max's weight variations (111.4 and 111.5) are negligible, and the wind conditions (TWA / TWS) are identical as they are interpolated from the coach's boat.

The raw mean SOG difference is small (−0.14 kn in favor of Chubanga) and not statistically significant. When separating by sailing regime, Levitaz is slightly faster upwind (0.107 kn) and Chubanga slightly faster downwind (0.409kn), but none of these differences are statistically significant.

2. Regression analysis: OLS on $\mathbf{SOG_{Levitaz} - SOG_{Chubanga}}$

a) All legs:

The model estimates the paired SOG difference between Levitaz and Chubanga, controlling sailing regime, master/leeward configuration, and which rider is sailing the Levitaz mast. The indicator function $\mathbf{1}(\cdot)$ equals one when the specified condition holds and zero otherwise:

$$\mathrm{SOG}_{\mathrm{Levitaz},i} - \mathrm{SOG}_{\mathrm{Chubanga},i} = \beta_0 + \beta_1 1(\mathrm{Upwind}_i) + \beta_2 1(\mathrm{Master}_i) + \beta_3 1(\mathrm{Max\ on\ Levi}_i) + \varepsilon_i$$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          d_avg_SOG_mast   R-squared:                       0.209
Model:                             OLS   Adj. R-squared:                  0.163
Method:                  Least Squares   F-statistic:                     5.482
Date:                 Wed, 07 Jan 2026   Prob (F-statistic):            0.00238
Time:                         15:37:04   Log-Likelihood:                -95.549
No. Observations:                   56   AIC:                             199.1
Df Residuals:                       52   BIC:                             207.2
Df Model:                            3
Covariance Type:                   HC3
====================================================================================================
                                       coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------------
Intercept                           -1.3400      0.466     -2.875      0.004      -2.254      -0.426
C(leg_type)[T.upwind]                0.5201      0.377      1.381      0.167      -0.218       1.258
C(boat1_master_leeward)[T.True]      0.4932      0.440      1.122      0.262      -0.368       1.355
C(levi_rider)[T.Max Maeder]          1.2483      0.410      3.046      0.002       0.445       2.052
==============================================================================
Omnibus:                        85.117   Durbin-Watson:                   1.803
Prob(Omnibus):                   0.000   Jarque-Bera (JB):             1521.692
Skew:                            4.161   Prob(JB):                         0.00
Kurtosis:                       27.143   Cond. No.                         4.05
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

The intercept represents the Levi–Chub SOG difference when Gian Stragiotti is sailing Levitaz, in downwind conditions and master role.

The intercept is negative and statistically significant (−1.34 kn, p = 0.004), indicating that when Gian Stragiotti sails Levitaz, Levi is on average 1.34 knots slower than Chubanga under the reference conditions. The rider coefficient is large and positive (+1.25 kn, p = 0.002), implying that when Max Maeder sails Levitaz, the Levi–Chub performance gap is reduced by about 1.25 knots, bringing it close to zero (-1.34+1.25 = -0.09 kn).

b) Upwind

$$\mathrm{SOG}_{\mathrm{Levitaz},i} - \mathrm{SOG}_{\mathrm{Chubanga},i} = \beta_0 + \beta_1 1(\mathrm{Master}_i) + \beta_2 1(\mathrm{Max\ on\ Levi}_i) + \varepsilon_i$$

```
======================================================================
Paired OLS — UPWIND ONLY
N paired intervals: 29
                            OLS Regression Results
==============================================================================
Dep. Variable:          d_avg_SOG_mast   R-squared:                       0.100
Model:                             OLS   Adj. R-squared:                  0.031
Method:                  Least Squares   F-statistic:                    0.6380
Date:                 Wed, 07 Jan 2026   Prob (F-statistic):              0.536
Time:                         15:37:04   Log-Likelihood:                -55.947
No. Observations:                   29   AIC:                             117.9
Df Residuals:                       26   BIC:                             122.0
Df Model:                            2
Covariance Type:                   HC3
====================================================================================================
                                       coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------------
Intercept                           -0.7480      0.461     -1.624      0.104      -1.651       0.155
C(boat1_master_leeward)[T.True]      0.8728      0.782      1.117      0.264      -0.659       2.405
C(levi_rider)[T.Max Maeder]          0.7858      0.710      1.107      0.268      -0.605       2.177
==============================================================================
Omnibus:                        60.403   Durbin-Watson:                   2.036
Prob(Omnibus):                   0.000   Jarque-Bera (JB):              495.714
Skew:                            4.227   Prob(JB):                    2.28e-108
Kurtosis:                       21.406   Cond. No.                         3.43
...

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

In the upwind-only regression, the intercept is negative (−0.75 kn) but not statistically significant (p = 0.104), indicating that Levi tends to be slower than Chub when Gian Stragiotti sails Levitaz, though this effect is not precisely identified. The rider coefficient is positive (+0.79 kn) but not statistically significant (p = 0.268), but this suggests again a smaller Levi–Chub gap when Max Maeder sails Levitaz, bringing the difference close to zero again.

Overall, upwind results are noisy and do not support clear conclusions on mast performance, but they are showing similar trends as the all legs case.

c) Downwind

$$\mathrm{SOG}_{\mathrm{Levitaz},i} - \mathrm{SOG}_{\mathrm{Chubanga},i} = \beta_0 + \beta_1 1(\mathrm{Master}_i) + \beta_2 1(\mathrm{Max\ on\ Levi}_i) + \varepsilon_i$$

```
======================================================================
Paired OLS — DOWNWIND ONLY
N paired intervals: 27
                            OLS Regression Results
==============================================================================
Dep. Variable:         d_avg_SOG_mast   R-squared:                       0.591
Model:                            OLS   Adj. R-squared:                  0.557
Method:                 Least Squares   F-statistic:                     15.49
Date:                Wed, 07 Jan 2026   Prob (F-statistic):           4.79e-05
Time:                        15:37:04   Log-Likelihood:                -28.793
No. Observations:                  27   AIC:                             63.59
Df Residuals:                      24   BIC:                             67.47
Df Model:                           2
Covariance Type:                  HC3
=====================================================================================================
                                    coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------------------
Intercept                        -1.4223      0.394     -3.614      0.000      -2.194      -0.651
C(boat1_master_leeward)[T.True]   0.1210      0.334      0.363      0.717      -0.533       0.775
C(levi_rider)[T.Max Maeder]       1.7189      0.350      4.914      0.000       1.033       2.405
==============================================================================
Omnibus:                        8.948   Durbin-Watson:                   1.948
Prob(Omnibus):                  0.011   Jarque-Bera (JB):                7.075
Skew:                          -1.105   Prob(JB):                       0.0291
Kurtosis:                       4.187   Cond. No.                         3.59
...

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

In the downwind-only regression, the intercept is negative and statistically significant (−1.42 kn, $p < 0.001$), indicating that when Gian Stragiotti sails Levitaz, Levi is on average 1.42 knots slower than Chubanga downwind. The rider coefficient is large and highly significant (+1.72 kn, $p < 0.001$), showing that when Max Maeder sails Levitaz, the Levi–Chub performance gap is strongly reduced and nearly eliminated. By contrast, the master/leeward configuration has no meaningful effect.

Overall, downwind performance differences are large, well identified, and primarily driven by rider-specific responses to the foil mast.

d) Conclusion:

The paired-difference analysis shows that Levi underperforms Chub when sailed by Gian Stragiotti, while this performance gap is largely eliminated when sailed by Max Maeder. There is no reversal of the gap: Max performs consistently well with both foils, whereas Gian performs noticeably better with Chubanga. This rider-specific pattern is weakly identified upwind due to higher variability, but emerges clearly and robustly downwind, where the Levi–Chub difference is large and statistically significant.

B. Approach II : Rider-Specific Analysis

The paired-difference analysis shows that rider identity plays a major role in the observed mast performance differences. This motivates a rider-specific analysis, which examines each rider separately to better understand how individual riders respond to the different foil masts.

1. Descriptive statistics and exploratory analysis

For Gian:

```
mast          Chub       Levi
leg_type
downwind  55.555556  44.444444
upwind    55.172414  44.827586
======================================================================
Gian – all legs
Average conditions – TWS: 17.043/16.200 kn, TWA: -59.195/-62.156°, Weight: 112.200/112.100 kg (Levi / Chub).
Mean SOG difference (Levi – Chub): -0.960 kn. Chub is faster on average in the raw data.
According to the Welch t-test on avg_SOG: t = -0.70, p = 0.489
The difference in mean SOG is not statistically significant (p = 0.489 ≥ 0.05).
======================================================================
Gian – upwind
Average conditions – TWS: 17.171/16.632 kn, TWA: 32.125/27.546°, Weight: 112.200/112.100 kg (Levi / Chub).
Mean SOG difference (Levi – Chub): -0.173 kn. Chub is faster on average in the raw data.
According to the Welch t-test on avg_SOG: t = -0.20, p = 0.844
The difference in mean SOG is not statistically significant (p = 0.844 ≥ 0.05).
======================================================================
Gian – downwind
Average conditions – TWS: 16.905/15.739 kn, TWA: -158.126/-157.839°, Weight: 112.200/112.100 kg (Levi / Chub).
Mean SOG difference (Levi – Chub): -1.730 kn. Chub is faster on average in the raw data.
According to the Welch t-test on avg_SOG: t = -4.10, p = 0.000
The difference in mean SOG is statistically significant (p = 0.000 < 0.05).
```

For Max:

```
mast          Chub       Levi
leg_type
downwind  44.444444  55.555556
upwind    44.827586  55.172414
======================================================================
Max – all legs
Average conditions – TWS: 16.200/17.043 kn, TWA: -61.620/-59.367°, Weight: 111.500/111.400 kg (Levi / Chub).
Mean SOG difference (Levi – Chub): 0.550 kn. Levi is faster on average in the raw data.
According to the Welch t-test on avg_SOG: t = 0.40, p = 0.693
The difference in mean SOG is not statistically significant (p = 0.693 ≥ 0.05).
======================================================================
Max – upwind
Average conditions – TWS: 16.632/17.171 kn, TWA: 28.689/31.500°, Weight: 111.500/111.400 kg (Levi / Chub).
Mean SOG difference (Levi – Chub): 0.315 kn. Levi is faster on average in the raw data.
According to the Welch t-test on avg_SOG: t = 0.72, p = 0.480
The difference in mean SOG is not statistically significant (p = 0.480 ≥ 0.05).
======================================================================
Max – downwind
Average conditions – TWS: 15.739/16.905 kn, TWA: -157.950/-157.806°, Weight: 111.500/111.400 kg (Levi / Chub).
Mean SOG difference (Levi – Chub): 0.723 kn. Levi is faster on average in the raw data.
According to the Welch t-test on avg_SOG: t = 2.38, p = 0.025
The difference in mean SOG is statistically significant (p = 0.025 < 0.05).
```

For both riders, mast exposure is well balanced across sailing regimes, with similar proportions of upwind and downwind intervals sailed on Levitaz and Chubanga. Average wind conditions and total weight are also comparable across masts within each rider, ensuring that observed differences are not driven by systematic condition imbalances.

For Gian Stragiotti, raw mean comparisons show that Chubanga consistently outperforms Levitaz. Across all legs, the mean SOG difference (Levi − Chub) is −0.96 kn, although this difference is not statistically significant. When separating by sailing regime, the gap is small and insignificant upwind (−0.17 kn), but becomes large and highly significant downwind (−1.73 kn, $p < 0.001$), indicating a substantial downwind performance advantage of Chubanga for Gian.

For Max Maeder, the pattern is reversed. Across all legs, Levitaz is faster on average (+0.55 kn), though this difference is not statistically significant. Upwind, the Levi advantage remains modest and insignificant (+0.32 kn). Downwind, however, Levitaz shows a clear and statistically significant advantage (+0.72 kn, $p = 0.025$), indicating better downwind performance for Max with Levitaz.

Overall, the rider-specific exploratory analysis reveals a strong contrast in rider–foil interactions: Gian performs significantly better downwind with Chubanga, while Max performs a bit better downwind with Levitaz. Upwind differences are small and not statistically significant for either rider. These patterns strongly suggest that foil mast performance depends on rider-specific responses, particularly in downwind conditions.

2. Regression analysis: OLS on SOG

a) Gian

Gian all legs:

$$\text{avg\_SOG}_i = \beta_0 + \beta_1\, 1(\text{Levi}_i) + \beta_2\, 1(\text{Upwind}_i) + \beta_3\, \text{TWS}_i + \beta_4\, 1(\text{Master}_i) + \varepsilon_i$$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                avg_SOG   R-squared:                       0.871
Model:                            OLS   Adj. R-squared:                  0.861
Method:                 Least Squares   F-statistic:                     147.5
Date:                Wed, 07 Jan 2026   Prob (F-statistic):           2.28e-27
Time:                        15:37:05   Log-Likelihood:                -113.97
No. Observations:                  56   AIC:                             237.9
Df Residuals:                      51   BIC:                             248.1
Df Model:                           4
Covariance Type:                  HC3
===============================================================================================
                                coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------------
Intercept                    33.9161      1.292     26.245      0.000      31.383      36.449
C(mast)[T.Levi]              -0.8023      0.535     -1.501      0.133      -1.850       0.245
C(leg_type)[T.upwind]        -9.4733      0.568    -16.671      0.000     -10.587      -8.360
C(master_leeward)[T.True]    -0.1348      0.629     -0.214      0.830      -1.367       1.098
avg_TWS                      -0.1209      0.085     -1.423      0.155      -0.288       0.046
==============================================================================
Omnibus:                       79.898   Durbin-Watson:                   1.534
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1109.802
Skew:                          -3.906   Prob(JB):                    1.02e-241
Kurtosis:                      23.362   Cond. No.                         86.0
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

Gian upwind:

$$\text{avg\_SOG}_i = \beta_0 + \beta_1\, \mathbf{1}(\text{Levi}_i) + \beta_2\, \text{TWS}_i + \beta_3\, \mathbf{1}(\text{Master}_i) + \varepsilon_i$$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                avg_SOG   R-squared:                       0.078
Model:                            OLS   Adj. R-squared:                 -0.032
Method:                 Least Squares   F-statistic:                     3.350
Date:                Wed, 07 Jan 2026   Prob (F-statistic):             0.0350
Time:                        15:37:05   Log-Likelihood:                -65.136
No. Observations:                  29   AIC:                             138.3
Df Residuals:                      25   BIC:                             143.7
Df Model:                           3
Covariance Type:                  HC3
===============================================================================================
                                coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------------
Intercept                    25.2167      2.534      9.952      0.000      20.250      30.183
C(mast)[T.Levi]              -0.0351      0.962     -0.037      0.971      -1.920       1.850
C(master_leeward)[T.True]    -0.4106      1.074     -0.382      0.702      -2.516       1.695
avg_TWS                      -0.1793      0.137     -1.310      0.190      -0.448       0.089
==============================================================================
Omnibus:                       53.414   Durbin-Watson:                   1.880
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              319.000
Skew:                          -3.720   Prob(JB):                     5.37e-70
Kurtosis:                      17.444   Cond. No.                         86.8
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

Gian downwind:

$$\text{avg\_SOG}_i = \beta_0 + \beta_1\,\mathbf{1}(\text{Levi}_i) + \beta_2\,\text{TWS}_i + \beta_3\,\mathbf{1}(\text{Master}_i) + \varepsilon_i$$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                avg_SOG   R-squared:                       0.418
Model:                            OLS   Adj. R-squared:                  0.342
Method:                 Least Squares   F-statistic:                     7.328
Date:                Wed, 07 Jan 2026   Prob (F-statistic):            0.00128
Time:                        15:37:05   Log-Likelihood:                -39.057
No. Observations:                  27   AIC:                             86.11
Df Residuals:                      23   BIC:                             91.30
Df Model:                           3
Covariance Type:                  HC3
================================================================================================
                               coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------------
Intercept                   32.8259      1.738     18.888      0.000      29.420      36.232
C(mast)[T.Levi]             -1.7176      0.467     -3.678      0.000      -2.633      -0.802
C(master_leeward)[T.True]    0.1767      0.592      0.299      0.765      -0.983       1.336
avg_TWS                     -0.0381      0.117     -0.324      0.746      -0.268       0.192
==============================================================================
Omnibus:                        5.552   Durbin-Watson:                   1.453
Prob(Omnibus):                  0.062   Jarque-Bera (JB):                3.665
Skew:                          -0.761   Prob(JB):                        0.160
Kurtosis:                       3.971   Cond. No.                         85.1
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

The reference category corresponds to Chubanga, non-master configuration, and Gian Stragiotti, and downwind conditions for the all legs analysis.

In the pooled regression, the Levitaz mast coefficient is negative (−0.80 kn) but not statistically significant ($p = 0.133$), indicating that after controlling for wind strength and sailing regime, the average Levi–Chub performance gap for Gian is not precisely identified. Sailing regime is the main driver of SOG, with a large and highly significant reduction in speed upwind relative to downwind.

Downwind, the OLS results are much clearer: the Levitaz mast is associated with a large and statistically significant SOG loss (−1.72 kn, $p < 0.001$), even after controlling for wind and configuration. This confirms that the downwind performance gap is a genuine mast effect for Gian, not driven by condition differences.

Upwind, the mast coefficient remains negative (-0.035 kn) but statistically insignificant ($p = 0.971$), indicating weak and poorly identified mast effects. Overall, OLS results confirm a strong, rider-specific downwind advantage of Chubanga for Gian, with limited evidence of meaningful mast effects upwind.

b) Max

Max all legs:

$$\text{avg\_SOG}_i = \beta_0 + \beta_1\,\mathbf{1}(\text{Levi}_i) + \beta_2\,\mathbf{1}(\text{Upwind}_i) + \beta_3\,\text{TWS}_i + \beta_4\,\mathbf{1}(\text{Master}_i) + \varepsilon_i$$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                avg_SOG   R-squared:                       0.969
Model:                            OLS   Adj. R-squared:                  0.967
Method:                 Least Squares   F-statistic:                     406.1
Date:                Wed, 07 Jan 2026   Prob (F-statistic):           5.47e-38
Time:                        15:37:05   Log-Likelihood:                -73.018
No. Observations:                  56   AIC:                             156.0
Df Residuals:                      51   BIC:                             166.2
Df Model:                           4
Covariance Type:                  HC3
===============================================================================================
                                coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------------
Intercept                    33.7592      0.780     43.280      0.000      32.230      35.288
C(mast)[T.Levi]               0.4587      0.268      1.709      0.087      -0.067       0.985
C(leg_type)[T.upwind]        -9.9258      0.260    -38.190      0.000     -10.435      -9.416
C(master_leeward)[T.True]    -0.2082      0.284     -0.732      0.464      -0.766       0.349
avg_TWS                      -0.0975      0.044     -2.208      0.027      -0.184      -0.011
==============================================================================
Omnibus:                       14.629   Durbin-Watson:                   1.772
Prob(Omnibus):                  0.001   Jarque-Bera (JB):               16.590
Skew:                          -1.096   Prob(JB):                     0.000250
Kurtosis:                       4.517   Cond. No.                         91.8
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

Max upwind:

$$\text{avg\_SOG}_i = \beta_0 + \beta_1\, \mathbf{1}(\text{Levi}_i) + \beta_2\, \text{TWS}_i + \beta_3\, \mathbf{1}(\text{Master}_i) + \varepsilon_i$$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                avg_SOG   R-squared:                       0.493
Model:                            OLS   Adj. R-squared:                  0.433
Method:                 Least Squares   F-statistic:                     6.354
Date:                Wed, 07 Jan 2026   Prob (F-statistic):            0.00237
Time:                        15:37:05   Log-Likelihood:                -34.046
No. Observations:                  29   AIC:                             76.09
Df Residuals:                      25   BIC:                             81.56
Df Model:                           3
Covariance Type:                  HC3
===============================================================================================
                                coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------------
Intercept                    25.8428      0.936     27.603      0.000      24.008      27.678
C(mast)[T.Levi]               0.2447      0.355      0.690      0.490      -0.450       0.940
C(master_leeward)[T.True]    -0.3933      0.363     -1.082      0.279      -1.105       0.319
avg_TWS                      -0.2039      0.057     -3.553      0.000      -0.316      -0.091
==============================================================================
Omnibus:                       26.880   Durbin-Watson:                   1.500
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               46.323
Skew:                          -2.145   Prob(JB):                     8.73e-11
Kurtosis:                       7.465   Cond. No.                         89.4
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

Max downwind:

$$\text{avg\_SOG}_i = \beta_0 + \beta_1\, \mathbf{1}(\text{Levi}_i) + \beta_2\, \text{TWS}_i + \beta_3\, \mathbf{1}(\text{Master}_i) + \varepsilon_i$$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                avg_SOG   R-squared:                       0.212
Model:                            OLS   Adj. R-squared:                  0.110
Method:                 Least Squares   F-statistic:                     2.446
Date:                Wed, 07 Jan 2026   Prob (F-statistic):             0.0896
Time:                        15:37:05   Log-Likelihood:                -30.832
No. Observations:                  27   AIC:                             69.66
Df Residuals:                      23   BIC:                             74.85
Df Model:                           3
Covariance Type:                  HC3
=============================================================================================
                                coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------------------
Intercept                    31.0462      1.091     28.450      0.000      28.907      33.185
C(mast)[T.Levi]               0.7662      0.352      2.174      0.030       0.075       1.457
C(master_leeward)[T.True]     0.0797      0.394      0.202      0.840      -0.693       0.852
avg_TWS                       0.0497      0.063      0.792      0.428      -0.073       0.173
==============================================================================
Omnibus:                        2.302   Durbin-Watson:                   1.345
Prob(Omnibus):                  0.316   Jarque-Bera (JB):                1.819
Skew:                          -0.485   Prob(JB):                        0.403
Kurtosis:                       2.178   Cond. No.                         94.6
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

In the pooled regression across all conditions, the coefficient on the Levitaz mast is positive (+0.46 kn) and marginally insignificant ($p = 0.087$), indicating that Max tends to sail slightly faster with Levitaz than with Chubanga once sailing regime, wind strength, and configuration are controlled for. As for Gian, sailing regime is the dominant driver of SOG, with a large and highly significant reduction in speed upwind relative to downwind.

Upwind, the Levitaz mast coefficient is small (+0.24 kn) and not statistically significant ($p = 0.49$), suggesting no meaningful difference in Max's upwind performance between the two foils. Wind strength remains the primary explanatory factor in this regime.

Downwind, by contrast, the Levitaz mast is associated with a positive and statistically significant SOG effect (+0.77 kn, $p = 0.030$). This indicates that Max sails significantly faster with Levitaz than with Chubanga downwind, even after controlling wind strength and configuration. Configuration effects remain negligible.

Overall, the OLS results show that Max's performance is largely insensitive to mast choice upwind, but that Levitaz provides a clear and statistically significant downwind advantage.

III. Conclusion

Both approaches gave us the same conclusions, foil mast performance is shown to depend strongly on rider-specific responses rather than on a uniform intrinsic advantage of one mast over the other. Gian exhibits a large and robust downwind advantage with Chubanga, whereas Max performs consistently well with both foils and shows a modest but statistically significant downwind advantage with Levitaz. Upwind effects are weak and imprecisely identified for both riders.